

Advanced 30-Day Backend Roadmap (Industry-Level)

This roadmap is designed to build strong backend fundamentals using an industry-standard stack. Each day includes: concept + 30 min project + upgrade version if you have extra time. Focus is on real backend thinking, not just small apps.

Industry Stack

Recommended Industry Stack: - Node.js + Express (or NestJS later) - TypeScript (VERY important for industry jobs) - MongoDB (start) → PostgreSQL (upgrade path) - Prisma ORM (modern DB tool) - JWT Authentication - Redis (caching basics) - Docker (deployment mindset) - Postman / Thunder Client - Git & GitHub (must) - Swagger (API documentation) - Deployment: Render / Vercel / Railway

Tools

Tools to Master: - VS Code + REST Client extensions - MongoDB Compass / pgAdmin - Postman - GitHub (version control) - Docker Desktop (later stage) - Railway / Render for deployment

Daily Roadmap

Day 1: Setup Express + TypeScript server. Learn how backend structure works. Build simple health check API. Upgrade: add environment variables.

Day 2: Routing system (GET/POST/PUT/DELETE). Build Notes API in memory. Upgrade: modular routes structure.

Day 3: Middleware concept (logger, request flow). Build request logger API. Upgrade: custom middleware chaining.

Day 4: Error handling system. Build safe calculator API. Upgrade: global error handler pattern.

Day 5: REST API design principles. Build Task API v1. Upgrade: clean REST naming conventions.

Day 6: Query params + filtering. Build product filter API. Upgrade: sorting + pagination logic.

Day 7: Mini project: Task Manager API without DB. Upgrade: architecture refactor.

Day 8: MongoDB connection + schema basics. Build user save API. Upgrade: schema validation.

Day 9: CRUD with database. Build Notes API with MongoDB. Upgrade: separate service layer.

Day 10: Schema design thinking. Build Todo system v2. Upgrade: relations concept.

Day 11: User-Task relationship. Upgrade version: nested data modeling.

Day 12: Pagination system. Upgrade: infinite scroll logic concept.

Day 13: Search feature implementation. Upgrade: text indexing concept.

Day 14: Full backend review project. Task Manager with DB.

Day 15: Password hashing (bcrypt). Build secure signup API.

Day 16: JWT authentication system. Build login API.

Day 17: Protected routes middleware. Build private notes API.

Day 18: Role-based access (admin/user). Upgrade: permission logic.

Day 19: Token handling & refresh concept.

Day 20: Validation (Zod/Joi). Build validated signup system.

Day 21: Complete Auth system project.

Day 22: File upload system (multer). Build image upload API.

Day 23: Environment config + security.

Day 24: Rate limiting concept. Build protected API.

Day 25: Caching basics (in-memory). Upgrade: Redis concept.

Day 26: Clean folder architecture (controllers/services/routes).

Day 27: Start Blog API project (real-world backend).

Day 28: Finish Blog API.

Day 29: Add authentication + roles to Blog API.

Day 30: Deploy project + GitHub portfolio cleanup.

Final Advice: If you only follow tutorials, you will stay average. Every day try to rebuild without help. Break your code intentionally and debug it. That is where real backend skill is built.